# Dr. Paveen Kalura

Project Scientist – II
Water Resources Systems Division
National Institute of Hydrology Roorkee
Contact: +91-9627988707
Email: praveenkalura@gmail.com


## EDUCATION

**Indian Institute of technology Roorkee** ***2019-2026***
*Doctor of Philosophy (Ph.D.)*

**National Institute of Technology, Warangal** ***2014-2016***
*Master of Technology (M. Tech.) in Remote Sensing and GIS*

**Graphic Era University, Dehradun** ***2008-2012***
*Bachelor of technology (B.Tech.) in Civil Engineering*

## EXPERIENCE

**Project Scientist II - National Institute of Hydrology, Roorkee (April 2025 – Present)**
***Project:*** *Indian Irrigation Management System*

- Updated frontend and backend of web based EQIPA tool being developed by the CWC and World Bank.
- Spatial database preparation for 10m Resolution data processing for version 2 of EQIPA tool.
- Developed In House Web Dashboard for GEE based EDIPA Tool to compute Indicators.

**Junior/Senior Research Fellow - Indian Institute of Technology Roorkee (March 2019 – February 2022)**
***Project: Water Resources Assessment under Climate Uncertainties (NRSC Funded)***

- Compiled and processed primary and secondary datasets related to climate, hydrology, land use, and groundwater for comprehensive water resources assessment of the Tons
- Applied hydrological modelling using the Soil and Water Assessment Tool (SWAT) to evaluate the impacts of climate variability and future climate scenarios on basin hydrology, evapotranspiration, and streamflow. Also Evaluated soil erosion risk and assessed Best Management Practices (BMPs)
- Prepared technical reports, conducted stakeholder workshop basin assessment documentation, and research manuscripts.

**Junior Research Fellow - *Indian Institute of Remote Sensing, Dehradun (August 2018 – March 2019)***
***Project: Soil Moisture Retrieval and Irrigated Area Mapping Using L- and S-band NISAR Data, and Assimilation in Land Surface Models***

- Assisted in development of methodology to use L- and S-band SAR datasets for retrieval of soil moisture and mapping of irrigated agricultural areas.
- Assisted in preparation of technical reports, research documentation related to the project outcomes.

**Assistant Professor - Graphic Era University, Dehradun (July 2016 – June 2017)**
***Department of Civil Engineering***

- Taught undergraduate courses in Surveying and conducted practical Surveying laboratory sessions, covering field surveying techniques, instruments, and data processing.
- Assisted students in practical GIS and surveying projects, including data preparation, mapping, and spatial analysis using GIS software.
- Taught Post Graduate courses in GIS and conducted GIS laboratory sessions.

**Intern - Indian Institute of Remote Sensing, Dehradun (July 2015 – June 2016)**
***Project: SARAL-AltiKa applications for inland waters, glaciers and Sunderban delta (SAC-ISRO sponsored SARAL-AltiKa science and utilization project)***

- Contributed to preparation of technical project reports and assisted in manuscript development related to the SARAL-AltiKa applications study.
- Processed and structured SARAL-AltiKa satellite altimetry data for analysis of inland water bodies and related hydrological applications.
- Developed GIS-ready data layers and spatial databases to facilitate analysis inland waters and deltaic regions.

## DISSERTATION/ THESIS WORK

### Ph.D. Dissertation

Synergetic application of VIC model and Water accounting plus for basin scale water resources

### M. Tech Thesis

Retrival of water level from SARAL/ALtiKA altimetry data and integration with hydrodynamic models

## INTERNSHIP/ CONSULTANCY PROJECT

**Consultancy Project – National Water Development Authority** **Mar 2022 - Sep 2023**

Project: Model Water Balance Study using system studies and Modelling Techniques for the Godavari basin between the Sri Ram Sagar project and Polavaram Project

**Research Internship - IIT Tirupati Navavishkar I-Hub Foundation** **Jan 2023 - Aug 2023**

Project: Estimating & Automatically Extracting Surface Water Extent and Chlorophyll Concentrations

## PUBLICATIONS

- **Kalura, P.,** Pandey, A., Dayal, D., & Chowdary, V. M. (2026). Enhancing Hydrologic Model Performance in a Data-Scarce Basin Using Satellite-Based Soil Moisture Data. ***Water Resources Management***, *40*(2), 79.
- Legese, B., **Kalura, P.,** Degife, A., Godana, G., & Chamara, J. (2025). Integrating GIS and Multi-Criteria Analysis for Sustainable Livestock Grazing: A Holistic Assessment of Biophysical and Socio-Economic Determinants in Semi-Arid Dawa Watershed of Ethiopia. ***Land Degradation & Development***, *36*(13), 4626-4642.
- **Kalura, P.,** Pandey, A., Chowdary, V. M., & Dayal, D. (2025). A TOPSIS-Based Multicriteria Assessment of Hydrologic Model Calibration Using Satellite-Derived Evapotranspiration and Streamflow Data. ***Hydrological Processes***, 39(7), e70191.
- **Kalura, P.,** Pandey, A., Chowdary, V. M., & Dayal, D. (2024). Evaluating hydrological responses of satellite precipitation products over an Indian tropical catchment through a distributed physical model. ***Hydrological Processes***, *38*(9), e15275.
- Dayal, D., Pandey, A., Gupta, P. K., & **Kalura, P.** (2024). Investigating the utility of satellite-based precipitation products for simulating extreme discharge events: an exhaustive model-driven approach for a tropical river basin in India. ***Environmental Monitoring and Assessment***, *196*(7), 608.
- Mudbhari, D., Kansal, M. L., & **Kalura, P.** (2022). Impact of climate change on water availability in Marsyangdi river basin, Nepal. ***Quarterly Journal of the Royal Meteorological Society***, *148*(744), 1407-1423.
- Swain, S., Mishra, S. K., Pandey, A., & **Kalura, P.** (2022). Inclusion of groundwater and socio-economic factors for assessing comprehensive drought vulnerability over Narmada River Basin, India: A geospatial approach. ***Applied Water Science***, *12*(2), 14.
- **Kalura, P.,** Pandey, A., Chowdary, V. M., & Raju, P. V. (2021). Assessment of hydrological drought vulnerability using geospatial techniques in the Tons River Basin, India. ***Journal of the Indian Society of Remote Sensing***, *49*(11), 2623-2637.
- Thakur, P. K., Garg, V., **Kalura, P.,** Agrawal, B., Sharma, V., Mohapatra, M., ... & Chauhan, P. (2021). Water level status of Indian reservoirs: A synoptic view from altimeter observations. ***Advances in Space Research***, *68*(2), 619-640.
- Himanshu, S. K., Singh, A. K., Kumar, S., & **Kalura, P.** (2013). Response of broccoli to irrigation scheduling and methods under drip, sprinkler and surface irrigation. ***International Journal of Engineering and Advanced Technology***, *2*(4), 777-782.

## CONFERENCE

- **Kalura, P.,** Pandey, A., Mishra, P. K., & Chowdary, V. M. (2024). Evaluating the Effects of Climate Change on Water Availability using Water accounting Plus (WA+) Framework in the Wardha River Basin: Insights from a Spatially Calibrated Hydrologic Model. 6th Workshop on Water Resources in Developing Countries: Hydroclimate Modeling, Information Tools and Simulation Techniques. (ICTP, Trieste, Italy)
- **Kalura, P.,** Pandey, A., Mishra, P. K., & Chowdary, V. M. (2024) Strategies for Transboundary Water Management: Remote Sensing based data insights into the Godavari Basin's resources using Water Accounting Plus (Wa+) framework. AGU Chapman Conference on Remote Sensing of the Water Cycle: Sensors to Science to Society-2024. Honolulu, Hawaii, USA

- **Kalura, P.,** Pandey, A., Mishra, P. K., & Chowdary, V. M.(2022) Water Accounting Plus (WA+) approach for integrated assessment of water resources" Water Accounting Plus (WA+) Symposium, Land and Water Management Department of IHE Delft, Netherlands.
- **Kalura, P.,** Pandey, A., Mishra, P. K., & Chowdary, V. M. (2023, December). Assessment of water budget component for improved decision making in a large river basins using Earth Observation Datasets: A Case Study from the Godavari River Basin, India. In AGU Fall Meeting Abstracts (Vol. 2023, No. 1741, pp. H31S-1741). Online
- **Kalura, P.,** Mishra, P. K., Pandey, A., Sankhua, R. N., & Chowdary, V.M. (2022, December). Assessment of Agricultural Water and Land Productivity Using Water Accounting Plus Framework for Pranhita River Basin. In AGU Fall Meeting Abstracts (Vol. 2022, pp. H25T-1347). Online
- **Kalura, P.,** Pandey, A., Chowdary, V. M., Dayal, D., Addressing In-Situ Data Scarcity in Hydrological Modelling: A Multivariate Calibration Approach Using Remote Sensing Datasets. Roorkee Water Conclave 2024 (RWC-2024), IIT Roorkee.
- **Kalura, P.,** Pandey, A., Chowdary, V. M., & Raju, P. V. (2022). A geospatial approach for spatiotemporal analysis and long-term trend detection in groundwater levels in the Tons river basin, India ISRS AND ISG Symposium, NRSC, Hyderabad, India
- "Indian inland water and parts of Antarctic ice sheet elevation and ice sheet velocity monitoring using altimetry and SAR based datasets". ISPRS TCV Mid-Term Symposium 2018, IIRS, Dehradun.
- "River water level and flow estimation using combination of space based altimeters and hydrodynamic models", Remote Sensing and Hydrology Symposium (ICRS-IAHS), Spain (2018).
- "Integration of Water Level Retrieved from Altimetry data with Hydrological Models" in the National Syposium on "Geomatics for Digital India" at JK Lakshmipat University, Jaipur (2015).
- "Local language based web information system for a village using open source geospatial tools" in the National Syposium on "Geomatics for Digital India" at JK Lakshmipat University, Jaipur (2015).
- "Development of Web Based Rural Utility System Using Open Source Geospatial Tool " June 2015, at FOSS4G 2015 - Second National Conference "Open Source Geospatial tools in climate change research and natural resources management" Dehradun (2015)

## SOFTWARE PROFICIENCY

- Operating Systems/Environments: Linux and Windows, High Performance Computing (HPC)
- Languages: Python and JavaScript
- Design and Analysis: ArcGIS, QGIS, GRASS GIS, Google Earth Engine
- Hydrological Models: SWAT, VIC and HEC-RAS
- Web Development: Django Framework, Node.Js, Geoserver, PostgreSQL

## GRANTS RECIEVED

- Received IIT Tirupati Navavishkar (NAV) I-Hub Foundation (ITTNiF) Research Grant under Navavishkar-in-Google Earth Engine (Nav-i-GEE) Program for project titled Estimating & Automatically Extracting Surface Water Extent and Chlorophyll Concentrations in Arakhvati river basin. (6 month – Research internship Program for IIT students)
- Received DST-SERB Internation Travel Scheme Grant by Science and Engineering Research Board (SERB), Department of Science and Technology (DST), under Ministry of Science and Technology, Government of India (File Number: ITS/2023/005560) to attend AGU chapman Conference in person at Honolulu, Hawaii., USA.
- Received International Centre for Theoretical Physics (ICTP) grant to visit 6th Workshop on Water Resources in Developing Countries: Hydroclimate Modeling, Information Tools and Simulation Techniques (20 - 31 May 2024) at Trieste, Italy.
- Received IHE Delft grant to attend WA+ Symposium, organised by IHE Delft, FAO and IWMI, held in Delft on July 12th and 13th 2022. Delft, Netherland

## PERSONAL INFORMATION

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|---|---|
| Name | Praveen Kalura |
| Gender | Male |
| Date of Birth | 07-02-1990 |
| Marital status | Married |
| Nationality | Indian |
| Permanent Address | H. No. 134, Shridev Suman Marg, Dhalwala,<br>Rishikesh, Uttarakhand, India Pin Code: 249201 |

## DECLARATION

I hereby solemnly declare that all the information mentioned above is true and correct according to my knowledge.

Dated: March 2026 (Praveen Kalura)